WIA2001 DATABASE

Assignment 1: Report

1. Introduction: Database Design for Ollie & Hana Puppy Nursery

Ollie & Hana Puppy Nursery is a growing **pet care brand** that focuses on providing safe, comfortable, and personalized services for dogs. The company currently operates two branches (Jaya One and United Point) and is in the process of opening **two more branches** at **River City** and **Good Choice.** In the long term, Ollie & Hana aims to expand into many neighbourhoods with smaller outlets, similar to the "Speedmart 99" or "7-Eleven" model, so that dog owners can easily access trusted pet care near their homes.

The nursery's core services are **boarding, grooming, and daycare**, which generate most of its daily revenue. In addition, Ollie & Hana also offers **dog training services** and runs a **small café area**, although these are currently secondary services. The business positions itself as a premium, homely environment where each dog's health, safety, and comfort are prioritised.

Operationally, most customer interactions begin online. New and existing customers usually contact the nursery through social media platforms such as Instagram, Facebook, and Xiaohongshu, or via WhatsApp. When a customer decides to use the service, they are directed to scan a **QR code** at the branch or from chat, which leads to an online registration form. In this form, the customer provides their details and **creates profiles** for their dogs, including name, age, breed, behaviour notes, and allergy or health information. All of this information is stored in a central system **shared across branches**.

Before a dog joins daycare or boarding, staff perform a basic **health and skin check** to look for issues like parasites or visible infections. Only dogs that pass this check are allowed into group play areas. During boarding or daycare, staff provide frequent updates to owners, such as **5–6 short video** clips per day, and some rooms are equipped with **live cameras** so that owners can view their dogs while they are in their assigned rooms. Boarding customers are also encouraged to bring their own food and bedding to match the dog's usual diet and sleeping habits, and to reduce the risk of allergy reactions.

As the business expands, Ollie & Hana faces increasing challenges in **managing data across multiple branches.** Bookings, pet profiles, health checks, loyalty points, and spending histories must be kept accurate and consistent so that any branch can serve a returning customer without confusion. Management also needs **clear reports** on **branch performance**, **service demand**, and **customer loyalty trends**. These needs highlight the importance of a **centralised database system** that can support day-to-day operations, ensure **data integrity,** and provide reliable information for future expansion decisions.

2. Business Rules (The Foundational Constraints)

The following **business rules** were identified based on interviews with Ollie & Hana Puppy Nursery staff and observations of their operational workflow. These rules define how data is structured, validated, and managed across all branches and will guide the development of the conceptual and logical database design.

2.1 Customer & Pet Registration Rules

- **A customer must register an account** through the QR-based online form before using any core service.

- **Each customer can register multiple pets**, but each pet must belong to **exactly one customer**.

- **Pet profiles must include required information** such as name, age, breed, behaviour notes, and **medical/allergy information**.

- **Customer and pet records are shared across all branches** through a centralized system.

2.2 Service Access & Operational Rules

- **A mandatory health and skin check must be performed** on every pet before entering daycare or boarding.

- **Only pets that pass the health check** may enter group play areas or shared spaces.

- **Boarding customers must bring their own food and bedding**, due to dietary preferences and allergy risks.

- **Service bundling is allowed**, such as “Boarding + Grooming” package deals.

- **Daily video updates (5–6 per day)** must be logged for boarding/daycare pets; **room cameras** may be accessible to customers when the pet is in its assigned room.

2.3 Branch & Employee Rules

- **Each branch has its own operating capacity**, including daycare capacity, boarding room capacity, and grooming slots.

- **Employees are assigned to exactly one branch**, but their system access roles determine what data they can view or modify.

- **Role-based access control (RBAC)** applies to all employees (e.g., Front Desk, Groomer, Pet Care Specialist, Branch Manager).

- **Only authorized employees** can update booking statuses, health checks, or financial transactions.

2.4 Booking & Payment Rules

- **A booking must be linked to at least one pet and one customer**.

- **A booking can contain multiple services**, such as daycare, grooming, training, or bundled packages.

- **Each service in a booking must have a corresponding price, staff assignment (if applicable), and timestamp.**

- **All completed services must be associated with a transaction**, which records payment amount, payment method, and points earned or redeemed.

- **A pet cannot have overlapping active bookings** (e.g., cannot be checked in twice on the same date/time).

2.5 Loyalty Program & Tier Rules

- **Customers earn points** based on spending; points can be **earned or redeemed** but **not both in the same transaction**.

- **Lifetime spending determines VIP tier**, following the company's rules (e.g., **OH Family < RM500**, **OH Lead > RM1000**).

- **Tier upgrades are automatic**, based on accumulated spending recorded in the database.

- **The points balance must always remain non-negative** and cannot drop below zero.

2.6 Centralised Multi-Branch Data Rules

- **All customer, pet, booking, and transaction data are stored in a centralised database**, accessible across all branches.

- **Branch-specific data**, such as capacity, employee assignments, and local service variations, is stored with foreign keys referencing the branch.

- **Reports must be able to aggregate data across all branches**, including sales, top services, and VIP trends.

3. User Requirements (System Objectives and Scope)

The following **user requirements** were identified based on interviews with staff members at Ollie & Hana Puppy Nursery. These requirements reflect what users need the system to store, maintain, track, and generate. They are grouped into three categories: **Data Definition & Maintenance**, **Searching & Tracking**, and **Reporting**.

3.1 Data Definition & Maintenance Requirements (by Staff & Management)

User Role	Requirement
Branch Administrator	**To maintain** (CREATE, UPDATE, DELETE) the details of all **Branch locations** and their associated contact information.
HR	**To maintain** all **Employee records**, including personnel details, assigned branch, system access **RoleID** (e.g., Groomer, Pet Care Specialist)**,**, and manager assignment.
Front Desk	**To maintain Customer Accounts** (personal details) and their linked **Pet Profiles** (health and dietary notes).**To maintain** the status of individual **Booking records** (e.g., Confirmed, Cancelled, Completed).
Branch Administrato	**To maintain** the catalogue of all available **Service Types**, their **standard pricing**, and **Bundle Definitions** (e.g., Boarding + Basic Grooming).

3.2 Searching & Tracking Requirements (by Staff)

User Role	Requirement
Front Desk	**To retrieve** a customer's **complete profile** by name, phone number, or pet name, including their current **Loyalty Tier** and **Points Balance**.
Front Desk	**To check** the availability of a specific service (e.g., Grooming Slot) at a specific **Branch** on a given **Date**.

Pet care assistant	**To track** and update a pet's real-time **Current Status** within the facility (e.g., In Daycare, In Boarding Room A, In Grooming).
Management	**To query** all **Pending/Overdue Payments** and generate a list of the associated customers.
Front Desk	**To process a Transaction** (sale and invoicing) which must simultaneously **update the Customer's Lifetime Spending** and calculate new **Points Accrual**.

3.3 Reporting Requirements (by Management)

User Role	Requirement
Executive Analyst	**To generate** a **Cross-Branch Performance Report** summarizing total revenue, most popular service, and number of new customer registrations **per branch** within a specified period (Complex JOIN + Aggregate).
Executive Analyst	**To identify** and list all **OH Lead customers** who have **not made any booking in the last 6 months**, and report their total Lifetime Spending and Points Balance. **(Complex JOIN + Temporal Nested Query).**(Objective: To support targeted Re-engagement Marketing campaigns by identifying high-value, inactive customers.)
Executive Analyst	**To report** on the **Top 5 most frequent Pet Allergies** recorded in the system, to guide inventory purchasing (Aggregate + JOIN).

Executive Analyst	**To perform a Service Dependency Analysis to determine the booking volume of Level 2 Training based on the completion rate of Level 1 Training. (Self-Referencing Query).**

4. Model the Data: Entities, Attributes, and Relationships

Based on the refined business rules and user requirements, the following core entities and their relationships are identified. This structure ensures high data integrity, system scalability, and the necessary complexity to support the multi-branch network and advanced reporting functionalities.

4.1. Core Entities and Attributes (Summary of Identified Tables)

Entity	Primary Key (PK)	Key Attributes (Critical Fields)	Foreign Keys (FK)	Rationale
BRANCH	BranchID	BranchName, Location, DaycareCapacity, BoardingCapacity, IsActive	None	Represents each physical nursery outlet; required to manage multi-branch operations.
EMPLOYEE	EmployeeID	FullName, Phone, HireDate, IsActive	RoleID: EMPLOYEE_ROLE; BranchID: BRANCH	Stores staff information and ensures every employee belongs to one branch and one role.
EMPLOYEE_ROLE	RoleID	RoleName, AccessLevel, Description	None	Defines role-based access control (RBAC) and system permissions.

CUSTOMER	CustomerID	FullName, Phone (Unique), Email, JoinDate, LifetimeSpending, PointsBalance	TierID: LOYALTY_TIER	Stores customer accounts shared across all branches; supports loyalty points and tier system.
LOYALTY_TIER	TierID	TierName, MinSpending, MaxSpending, BenefitNotes	None	Defines VIP tiers used for customer loyalty and spending classification.
PET	PetID	PetName, Breed, Species, DateOfBirth/Age, AllergyNotes, BehaviourNotes, Status	CustomerID: CUSTOMER	Represents each registered pet; essential for bookings, health checks, and medical logs.
HEALTH_CHECK	CheckID	CheckDateTime, Result, Notes	PetID: PET; EmployeeID: EMPLOYEE	Mandatory pre-boarding/daycare health inspections to ensure safety.
PET_MEDICAL_LOG	MedicalLogID	LogDateTime, IssueType, Notes	PetID: PET; RecordedByEmployeeID: EMPLOYEE	Tracks ongoing medical or skin issues for each pet.
SERVICE_TYPE	ServiceID	ServiceName, Category, DefaultDuration, BasePrice, IsBundle	None	Defines standard services and bundles offered by the nursery.
BOOKING	BookingID	BookingDateTime, StartDateTime, EndDateTime, Channel, Status	CustomerID :CUSTOMER; BranchID: BRANCH	Customer booking for services at a specific branch and time.

BOOKING_PET	BookingID, PetID	SpecialInstruction	BookingID: BOOKING; PetID: PET	Associative table enabling multi-pet bookings; resolves M:N relationship.
BOOKING_SERVICE_LINE	BookingLineID	Quantity, LinePrice, ServiceStatus	BookingID: BOOKING; ServiceID: SERVICE_TYPE; AssignedEmployeeID: EMPLOYEE	Lists all individual services included in a booking.
TRANSACTION	TransactionID	TransactionDateTime, TotalAmount, PaymentMethod, PointsEarned, PointsRedeemed	CustomerID: CUSTOMER; BranchID: BRANCH	Represents payment for completed services; used for loyalty tracking.
TRANSACTION_SERVICE_LINE	TxnLineID	Quantity, LineAmount	TransactionID: TRANSACTION; ServiceID: SERVICE_TYPE	Provides detailed breakdown of services paid in a transaction.

4.2. Relationship Definitions and M:N Resolution

Relationship	Keys Involved	Cardinality	Participation	Rationale
BRANCH – EMPLOYEE	BranchID (PK) - BranchID (FK)	1 BRANCH: M EMPLOYEES	Mandatory on EMPLOYEE	Each employee must belong to exactly one physical branch.

EMPLOYEE_ROLE – EMPLOYEE	RoleID (PK) - RoleID (FK)	1 ROLE: M EMPLOYEES	Mandatory on EMPLOYEE	Ensures role-based access control (RBAC).
CUSTOMER – PET	CustomerID (PK) - CustomerID (FK)	1 CUSTOMER : M PETS	Mandatory on PET	One customer can register several pets; each pet has exactly one owner.
LOYALTY_TIER – CUSTOMER	TierID (PK) - TierID (FK)	1 TIER: M CUSTOMERS	Optional on CUSTOMER	A customer may or may not have a tier; tiers classify customers by spending.
PET – HEALTH_CHECK	PetID (PK) - PetID (FK)	1 PET: M HEALTH_CHECKS	Mandatory on HEALTH_CHECK	Logs mandatory health checks before daycare/boarding.
EMPLOYEE – HEALTH_CHECK	EmployeeID (PK) - EmployeeID (FK)	1 EMPLOYEE: M HEALTH CHECKS	Mandatory on HEALTH_CHECK	Identifies which staff performed the check.
PET – PET_MEDICAL_LOG	PetID (PK) - PetID (FK)	1 PET: M MEDICAL_LOGS	Optional on PET	Tracks ongoing medical/skin issues for each pet.
CUSTOMER – BOOKING	CustomerID (PK) - CustomerID (FK)	1 CUSTOMER : M BOOKINGS	Mandatory on BOOKING	Bookings must be made under a customer account.
BRANCH – BOOKING	BranchID (PK) - BranchID (FK)	1 BRANCH: M BOOKINGS	Mandatory on BOOKING	Bookings occur at a specific branch.
BOOKING – PET (via BOOKING_PET)	BookingID, PetID (Composite PK)	M: N (1 BOOKING: M PETS; 1 PET: M BOOKINGS)	Mandatory for both	Allows multi-pet bookings and historical booking records.
BOOKING – BOOKING_SERVICE_LINE	BookingID (PK) - BookingID (FK)	1 BOOKING: M BOOKING_SERVICE_LINES	Mandatory on SERVICE_LINE	Each booking may contain multiple service items.
SERVICE_TYPE –	ServiceID (PK) -	1 SERVICE_TYPE :	Mandatory on	Specifies which services

BOOKING_SERVICE_LINE	ServiceID (FK)	M BOOKING_SERVICE_LINES	SERVICE_LINE	were booked.
CUSTOMER – TRANSACTION	CustomerID (PK) - CustomerID (FK)	1 CUSTOMER : M TRANSACTIONS	Mandatory on TRANSACTION	All payments belong to a customer account.
BRANCH – TRANSACTION	BranchID (PK) - BranchID (FK)	1 BRANCH: M TRANSACTIONS	Mandatory on TRANSACTION	Transactions occur at a branch for reporting purposes.
TRANSACTION – TRANSACTION_SERVICE_LINE	TransactionID (PK) - TransactionID (FK)	1 TRANSACTION: M TXN_LINES	Mandatory on TXN_LINE	Each transaction contains detailed service items.
SERVICE_TYPE – TRANSACTION_SERVICE_LINE	ServiceID (PK) - ServiceID (FK)	1 SERVICE_TYPE : M TXN_LINES	Mandatory on TXN_LINE	Specifies which services were paid for.

4.3 Integrity and Business Constraints

This section describes the integrity rules and business constraints that must be enforced by the Ollie & Hana Puppy Nursery database to ensure data accuracy, consistency, and alignment with real-world operations.

4.3.1 Key and Entity Integrity Constraints

1. Every table has a primary key (PK) that uniquely identifies each record:

    a. BRANCH (BranchID), EMPLOYEE (EmployeeID), EMPLOYEE_ROLE (RoleID), CUSTOMER (CustomerID), LOYALTY_TIER (TierID), PET (PetID), HEALTH_CHECK (CheckID), PET_MEDICAL_LOG (MedicalLogID), SERVICE_TYPE (ServiceID), BOOKING (BookingID), BOOKING_SERVICE_LINE (BookingLineID), TRANSACTION (TransactionID), TRANSACTION_SERVICE_LINE (TxnLineID).

    b. BOOKING_PET has a composite primary key (BookingID, PetID).

2. All primary key attributes must be non-null.

    a. No row can exist without a valid PK value.

3. The customer phone number is unique.

    a. CUSTOMER.Phone must not contain duplicates, to avoid multiple accounts using the same number.

4. Status-related fields use controlled values.

    a. Examples: Booking status (Pending, Confirmed, Checked-In, Completed, Cancelled), Health check result (Pass, Fail), ServiceStatus (Pending, In Progress, Completed).

4.3.2 Referential Integrity Constraints

1. Employees must be linked to existing branches and roles.

    - EMPLOYEE.BranchID references BRANCH.BranchID.

    - EMPLOYEE.RoleID references EMPLOYEE_ROLE.RoleID.

2. Pets must be linked to existing customers.

    - PET.CustomerID references CUSTOMER.CustomerID.

3. Customer tiers must be valid if present.

    - CUSTOMER.TierID (if not null) references LOYALTY_TIER.TierID.

4. Health checks and medical logs must reference valid pets and employees.

    - HEALTH_CHECK.PetID references PET.PetID.

    - HEALTH_CHECK.EmployeeID references EMPLOYEE.EmployeeID.

- PET_MEDICAL_LOG.PetID references PET.PetID.
- PET_MEDICAL_LOG.RecordedByEmployeeID references EMPLOYEE.EmployeeID.

5. Bookings must be linked to valid customers and branches.
    - BOOKING.CustomerID references CUSTOMER.CustomerID.
    - BOOKING.BranchID references BRANCH.BranchID.
6. Booking–pet links must reference existing bookings and pets.
    - BOOKING_PET.BookingID references BOOKING.BookingID.
    - BOOKING_PET.PetID references PET.PetID.
7. Booking service lines must reference existing bookings, services and employees.
    - BOOKING_SERVICE_LINE.BookingID references BOOKING.BookingID.
    - BOOKING_SERVICE_LINE.ServiceID references SERVICE_TYPE.ServiceID.
    - BOOKING_SERVICE_LINE.AssignedEmployeeID (if not null) references EMPLOYEE.EmployeeID.
8. Transactions must be linked to valid customers and branches.
    - TRANSACTION.CustomerID references CUSTOMER.CustomerID.
    - TRANSACTION.BranchID references BRANCH.BranchID.
9. Transaction service lines must reference existing transactions and services.
    - TRANSACTION_SERVICE_LINE.TransactionID references TRANSACTION.TransactionID.
    - TRANSACTION_SERVICE_LINE.ServiceID references SERVICE_TYPE.ServiceID.

4.3.3 Business Rule Constraints

1. Non-overlapping active bookings per pet

    - A pet cannot have two overlapping active bookings (e.g. Confirmed or Checked-In) for the same date and time range in BOOKING_PET.

2. Mandatory health check before daycare or boarding

    - Before a pet is marked as Checked-In for daycare or boarding, there must be a recent HEALTH_CHECK record with Result = 'Pass' for that pet.

3. Branch capacity limits

    - The total number of active boarding bookings for a given time period at a branch must not exceed BRANCH.BoardingCapacity.

    - The total number of active daycare bookings for a given time period at a branch must not exceed BRANCH.DaycareCapacity.

4. Loyalty points and balance rules

    - PointsBalance for a customer must never be negative.

    - PointsRedeemed in a transaction cannot be greater than the customer's current PointsBalance.

    - The system must enforce valid use of points (for example, points cannot be both heavily earned and redeemed in a way that breaks the business rule in a single transaction).

5. Tier and lifetime spending consistency

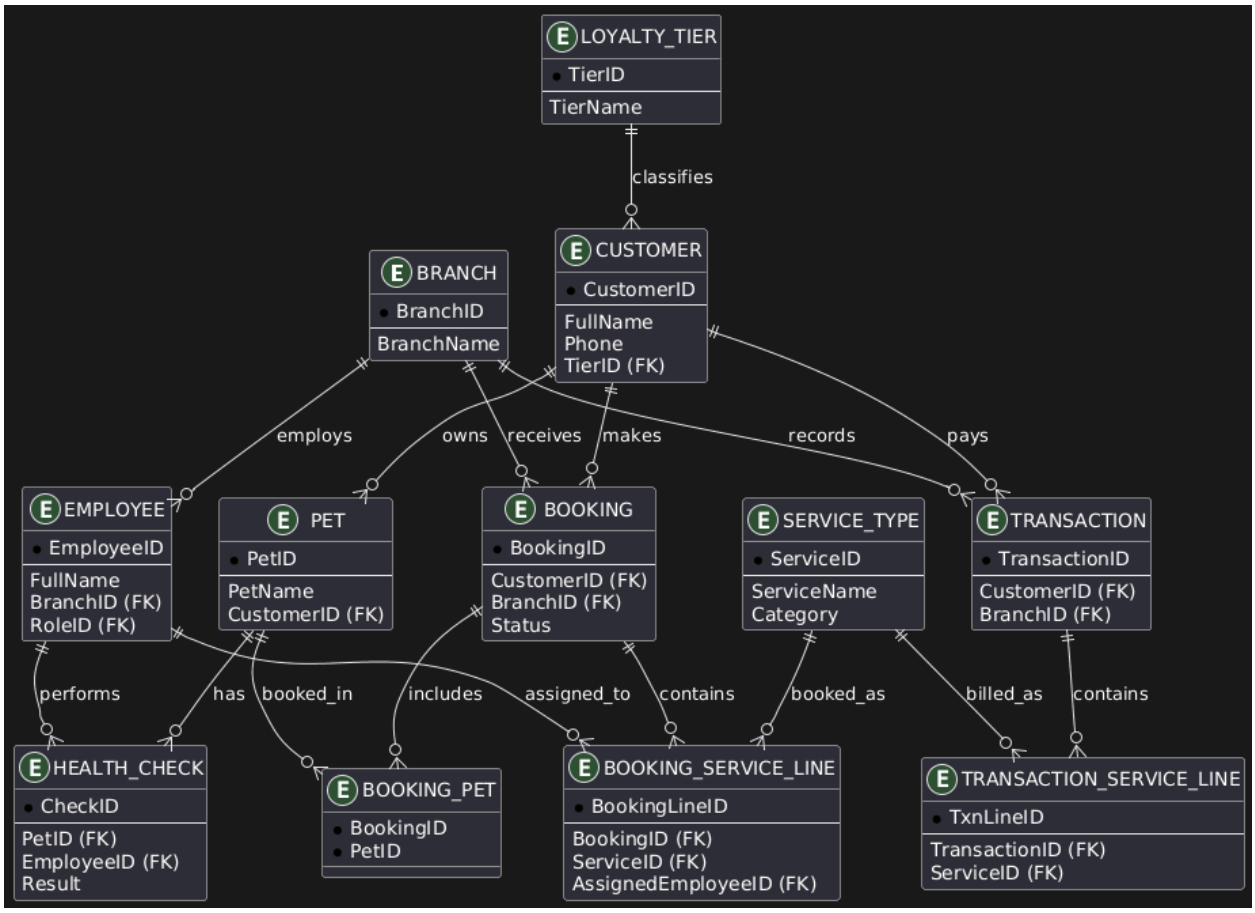
    - CUSTOMER.LifetimeSpending must be consistent with the tier ranges defined in LOYALTY_TIER (MinSpending, MaxSpending).

    - When a customer's LifetimeSpending crosses a threshold, their TierID should be updated to the correct tier.

6. Service and booking consistency

- Every BOOKING must have at least one related BOOKING_SERVICE_LINE.
- Every TRANSACTION should have at least one transaction service line to represent what was paid for.

5. Draw ERD (Final Normalized ERD)



6.Normalize the Tables (1NF → 2NF → 3NF)

This section demonstrates the normalization process for the Booking subsystem.The booking process was chosen because it contains multiple repeating groups (pets, services, employees), making it suitable for illustrating the UNF → 1NF → 2NF → 3NF transformation.

6.1 Unnormalized Form (UNF)

In the unnormalized state, booking information may be stored as one large table:

BookingID	CustomerName	Customer Phone	BranchName	PetName	Service Name	Service Price	AssignedEmployee	Booking Date	Status
B001	John Tan	12345	Jaya One	Coco	Grooming	75	Alice	2025-11-10	Confirmed
B001	John Tan	12345	Jaya One	Coco	Daycare	40	Ben	2025-11-10	Confirmed
B001	John Tan	12345	Jaya One	Milo	Daycare	40	Ben	2025-11-10	Confirmed

Problems in UNF:

1. Repeating customer details
2. Repeating pet details
3. Repeating branch details
4. Multiple services inside one booking
5. Multi-valued attributes for services and pets
6. High redundancy → update, insert and delete anomalies

6.2 First Normal Form (1NF)

Rule:

1. No repeating groups
2. All attributes must be atomic

To reach 1NF, we break each service/pet group into individual rows:

BookingID	CustomerID	BranchID	PetID	ServiceID	AssignedEmployeeID	BookingDate	Status
B001	C001	BR01	P001	S001	E003	2025-11-10	Confirmed
B001	C001	BR01	P001	S002	E002	2025-11-10	Confirmed

B001	C001	BR01	P002	S002	E002	2025-11-10	Confirmed

Remaining issues:

1. Composite primary key (BookingID, PetID, ServiceID)
2. Attributes depend on parts of the key (partial dependency)
3. Still contains transitive dependencies (CustomerID → CustomerName, Phone)

6.3 Second Normal Form (2NF)

Rule:

- Must be in 1NF
- No partial dependencies (non-key attribute depending on only part of a composite key)

Composite Key in 1NF:
(BookingID, PetID, ServiceID)

We identify partial dependencies:

1. BookingDate, Status → depend on BookingID only
2. PetName → depends on PetID only
3. ServiceName, BasePrice → depend on ServiceID only
4. EmployeeName → depends on EmployeeID only

Therefore, we split the 1NF table into modular tables:

a) BOOKING

BookingID (PK) | CustomerID | BranchID | BookingDate | Status |

b) BOOKING_PET

BookingID (PK, FK) | PetID (PK, FK) |

c) BOOKING_SERVICE_LINE

BookingLineID (PK) | BookingID (FK) | ServiceID (FK) | AssignedEmployeeID (FK) |

d) PET

PetID (PK) | CustomerID (FK) | PetName | Breed | AllergyNotes | BehaviourNotes |

e) SERVICE_TYPE

ServiceID (PK) | ServiceName | Category | BasePrice |

At this stage:

- Every non-key attribute fully depends on its key
- Partial dependencies removed → **2NF achieved**

6.4 Third Normal Form (3NF)

Rule:

- Must be in 2NF
- Remove transitive dependencies (non-key → non-key)

Identify transitive dependencies:

1. CustomerID → CustomerName, Phone, TierID (move to CUSTOMER table)
2. BranchID → BranchName (move to BRANCH table)
3. EmployeeID → FullName, RoleID, BranchID (move to EMPLOYEE table)
4. ServiceID → Category, BasePrice (already separated)

Final 3NF tables:

BOOKING

BookingID | CustomerID | BranchID | BookingDate | Status |

BOOKING_PET

BookingID | PetID |

BOOKING_SERVICE_LINE

BookingLineID | BookingID | ServiceID | AssignedEmployeeID |

CUSTOMER

CustomerID | FullName | Phone | TierID |

PET

PetID | CustomerID | PetName | Species | Breed |

SERVICE_TYPE

ServiceID | ServiceName | Category | BasePrice |

BRANCH

BranchID | BranchName | Location |

EMPLOYEE

EmployeeID | FullName | RoleID | BranchID |

TRANSACTION

TransactionID | CustomerID | BranchID | TotalAmount | PointsEarned | PointsRedeemed |

TRANSACTION_SERVICE_LINE TxnLineID | TransactionID | ServiceID | LineAmount |

1. **No partial dependencies**
2. **No transitive dependencies**
3. **No repeating groups**
4. **All tables correspond to the final ERD**

Normalization is complete.

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7. Final ERD (After Normalization)

Since all tables are already in 3NF and the ERD was built from normalized tables,
the final ERD remains the same as the one produced in Step 5.